EEME 30004
Communications & Networks

RF Components #1

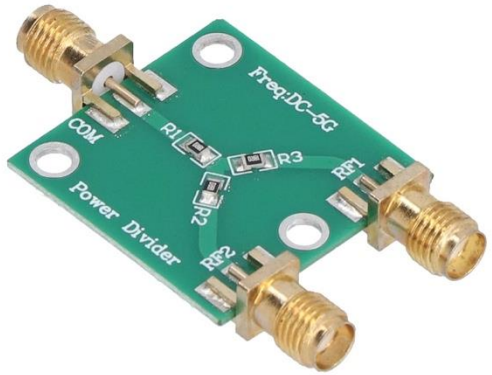
Dr Sean Gu

Passive vs Active Components

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Passive RF Components

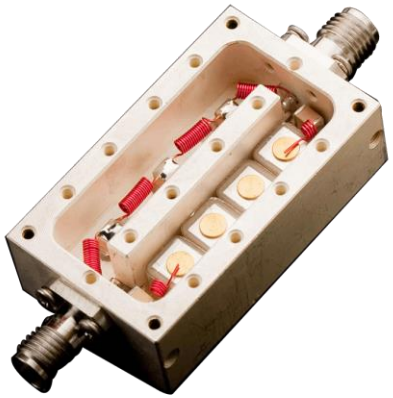
- Definition: Operate without external power, relying on physical properties to process signals.
- Key Features:
  - No signal amplification;
  - Transfer energy via electromagnetic fields;
  - Linear operation (e.g., impedance matching);
  - Examples: Resistors, capacitors, inductors, filters, antennas.



power dividers



couplers



filters



Passive vs Active Components

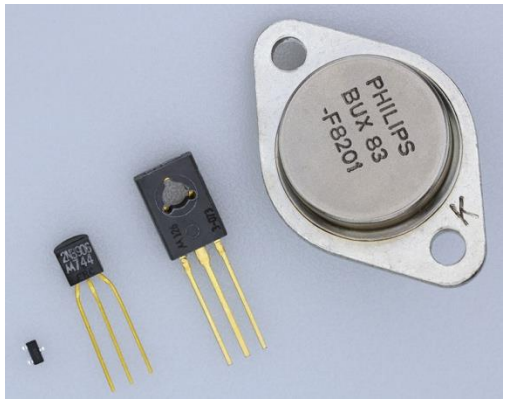
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Active RF Components

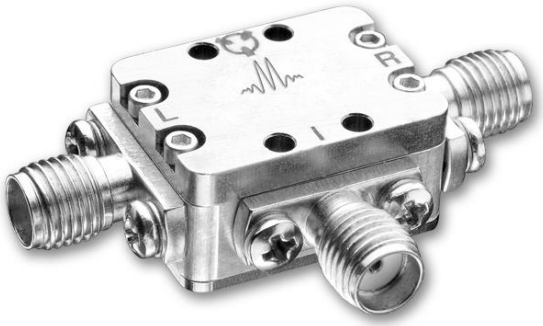
- Definition: require external power to amplify or modulate signals
- Key features:
  - Depend on external power (e.g., DC supply);
  - Provide signal gain (e.g., amplifiers);
  - Enable frequency conversion (e.g., mixers);
  - Examples: Transistors, RF amplifiers, oscillators, active filters.



amplifiers



transistors



mixers



oscillators

Passive vs Active Components

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Aspect	Active Components	Passive Components
Power Dependency	Require external power	No power needed
Signal Handling	Amplify/generate signals	Transmit/attenuate signals
Noise Level	May introduce noise (e.g., thermal)	Ultra-low noise
Power Consumption	Higher	Negligible
Cost	Higher (complex semiconductors)	Lower (simple structure)

Active Component Use Cases

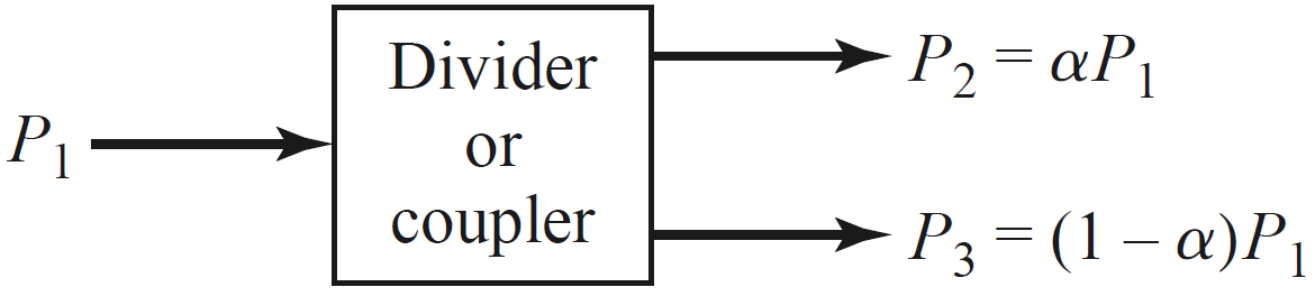
- Signal Amplification: 5G base station power amplifiers (PAs);
- Frequency Generation: Local oscillators (LOs) for carrier waves;
- Modulation/Demodulation: Mixers in RF transceivers.

Passive Component Use Cases

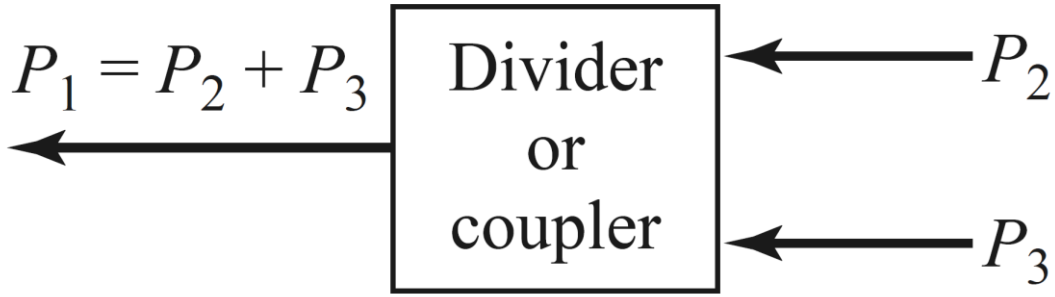
- Impedance Matching: Baluns in antenna feedlines;
- Filtering: Bandpass filters in Wi-Fi modules;
- Power Distribution: Power dividers.

Power Divider / Coupler

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power division



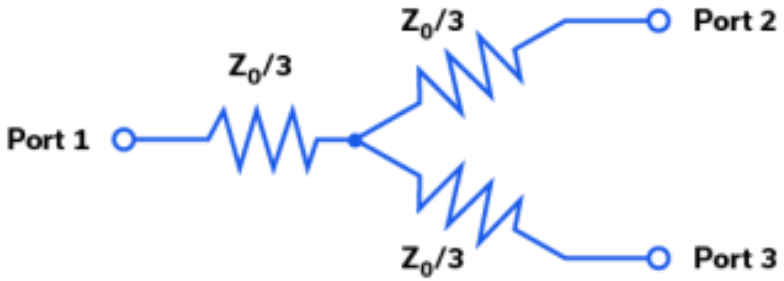
power combination

Power Divider

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A 3-port ***cannot*** achieve lossless, reciprocal, and matched at all ports at the same time.

resistive divider: matched at all ports, reciprocal but lossy

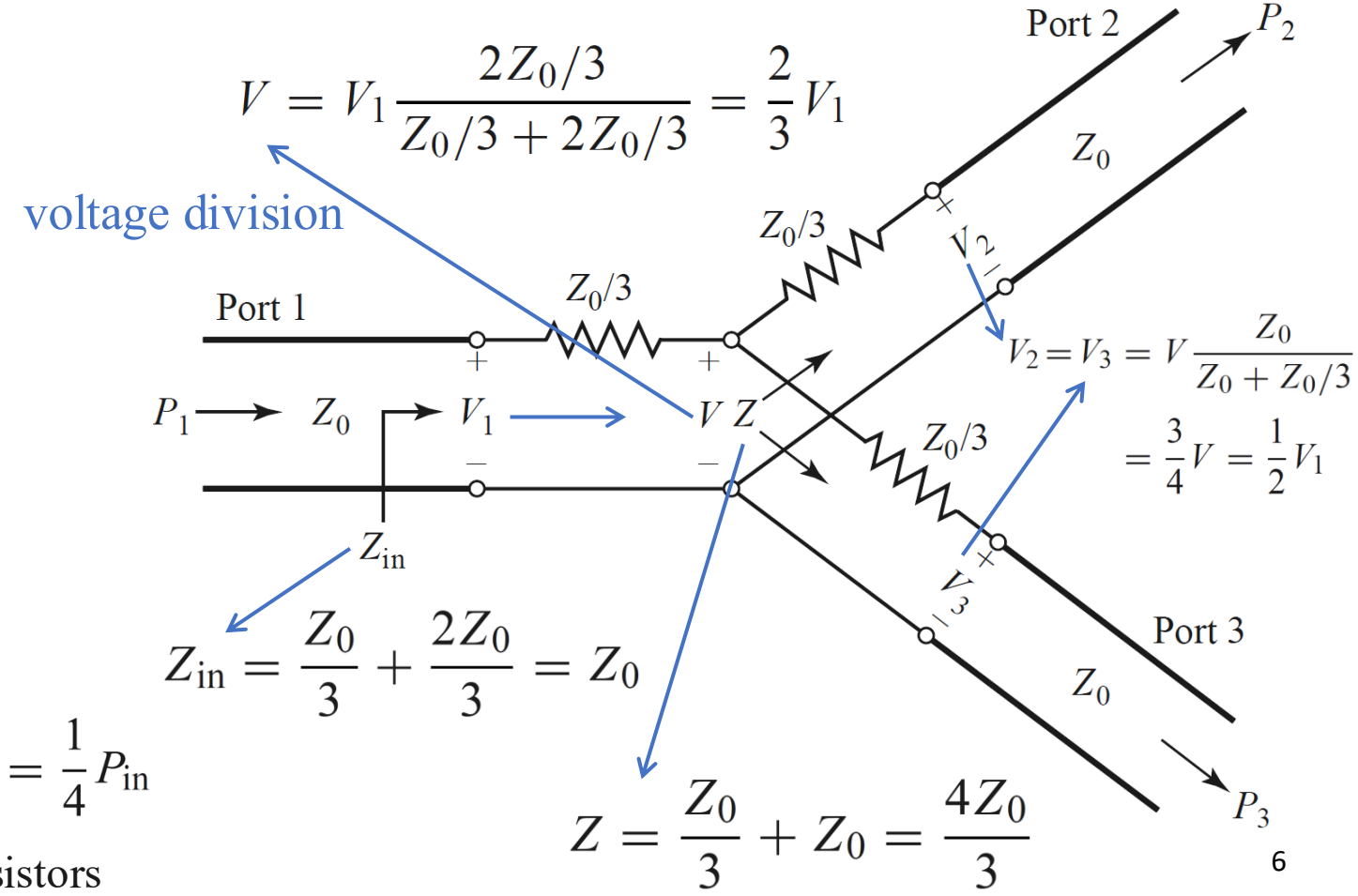


$$[S] = \frac{1}{2}\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

at port 1: $P_{\text{in}} = \frac{1}{2}\frac{V_1^2}{Z_0}$

at ports 2&3: $P_2 = P_3 = \frac{1}{2}\frac{(1/2V_1)^2}{Z_0} = \frac{1}{8}\frac{V_1^2}{Z_0} = \frac{1}{4}P_{\text{in}}$

half of the supplied power is dissipated in the resistors

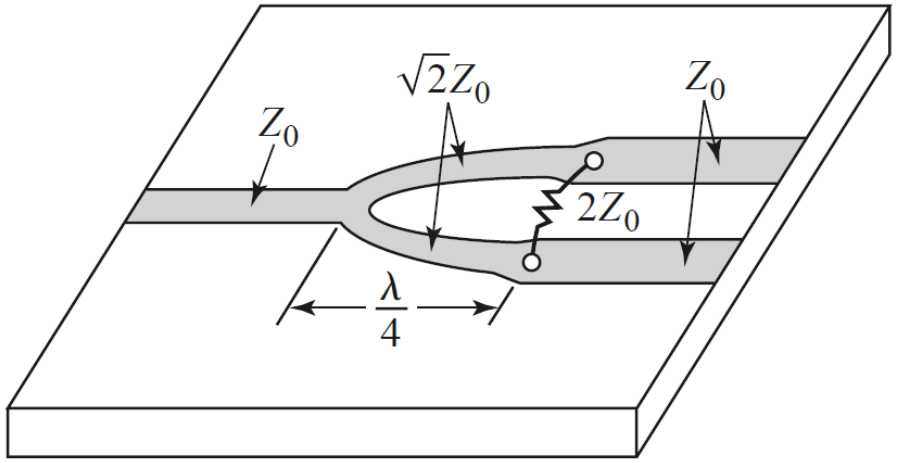


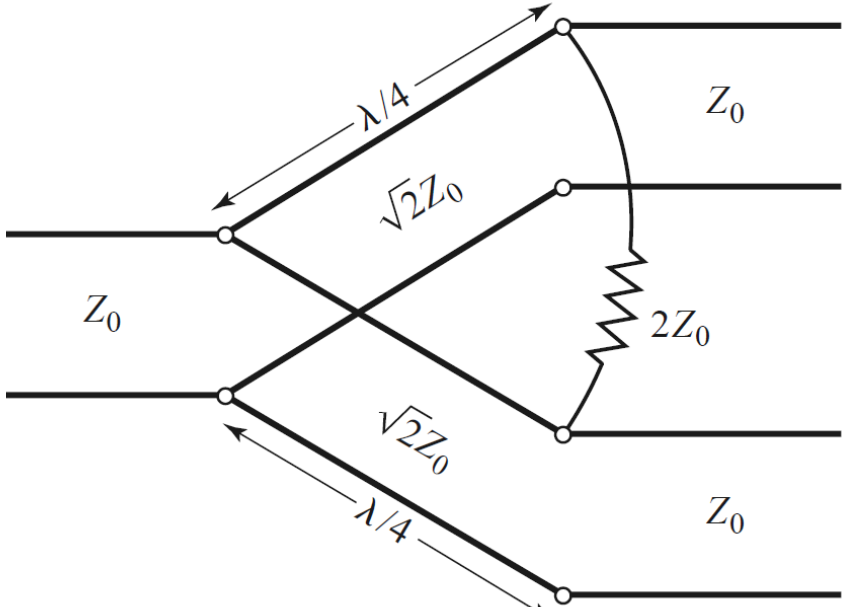
Power Divider

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A 3-port ***cannot*** achieve lossless, reciprocal, and matched at all ports at the same time.

Wilkinson divider: matched at all ports, reciprocal but lossy





$$[S] = \begin{bmatrix} 0 & -\frac{j}{\sqrt{2}} & -\frac{j}{\sqrt{2}} \\ -\frac{j}{\sqrt{2}} & 0 & 0 \\ -\frac{j}{\sqrt{2}} & 0 & 0 \end{bmatrix}$$

$S_{11} = 0$ ($Z_{\text{in}} = 1$ at port 1)

$S_{22} = S_{33} = 0$ (ports 2 and 3 matched for even and odd modes)

$S_{12} = S_{21} = \dfrac{V_1^e + V_1^o}{V_2^e + V_2^o} = -j/\sqrt{2}$ (symmetry due to reciprocity)

$S_{13} = S_{31} = -j/\sqrt{2}$ (symmetry of ports 2 and 3)

$S_{23} = S_{32} = 0$ (due to short or open at bisection)



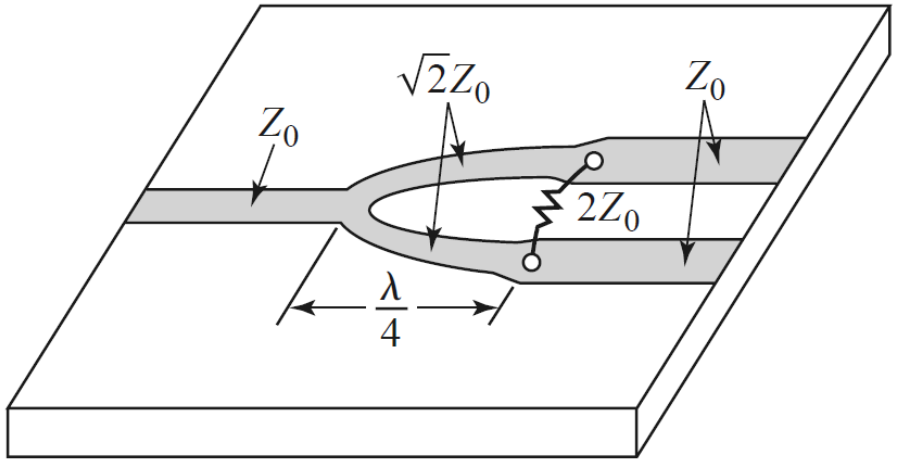
Why is it lossy?

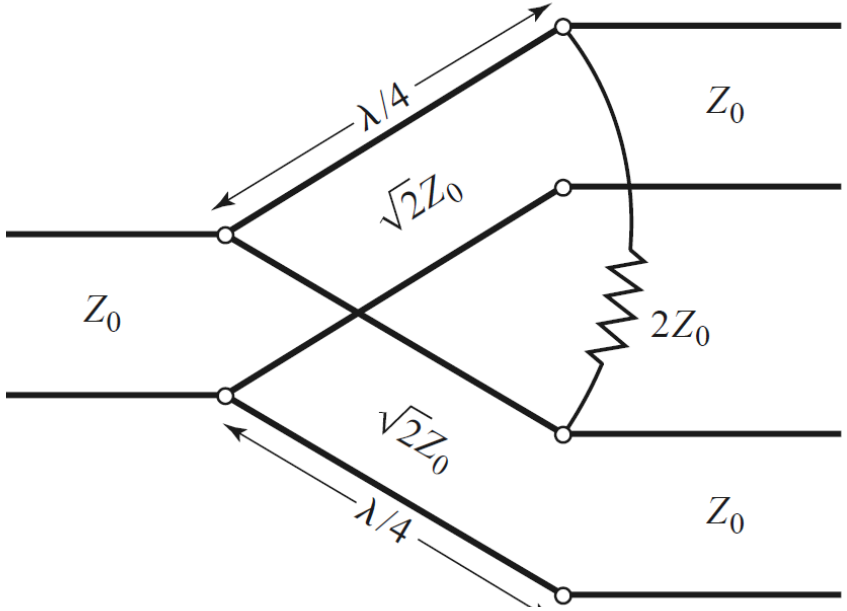
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$S_{23} = S_{32} = 0$ (due to short or open at bisection)

If input at port 1, and outputs are matched, no power is dissipated in the resistor.

Only reflected power from/or input power from ports 2 or 3 is dissipated in the resistor.

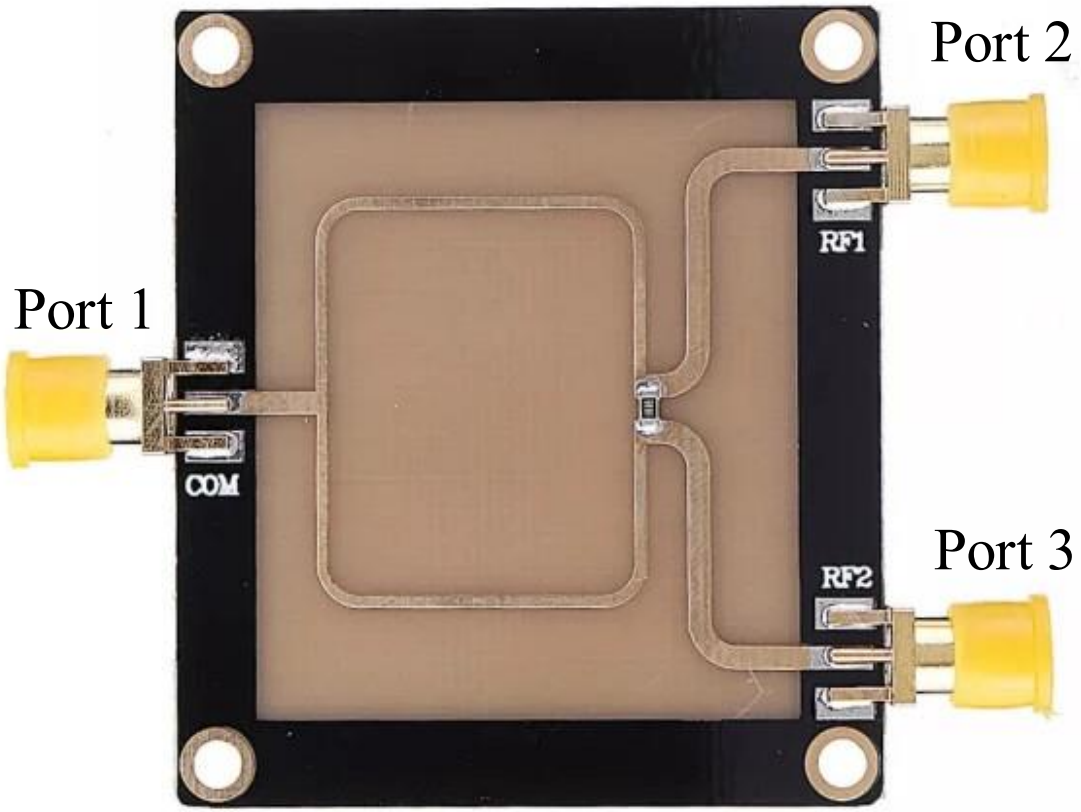
$S_{23} = S_{32} = 0$ indicates ports 2 and 3 are isolated.

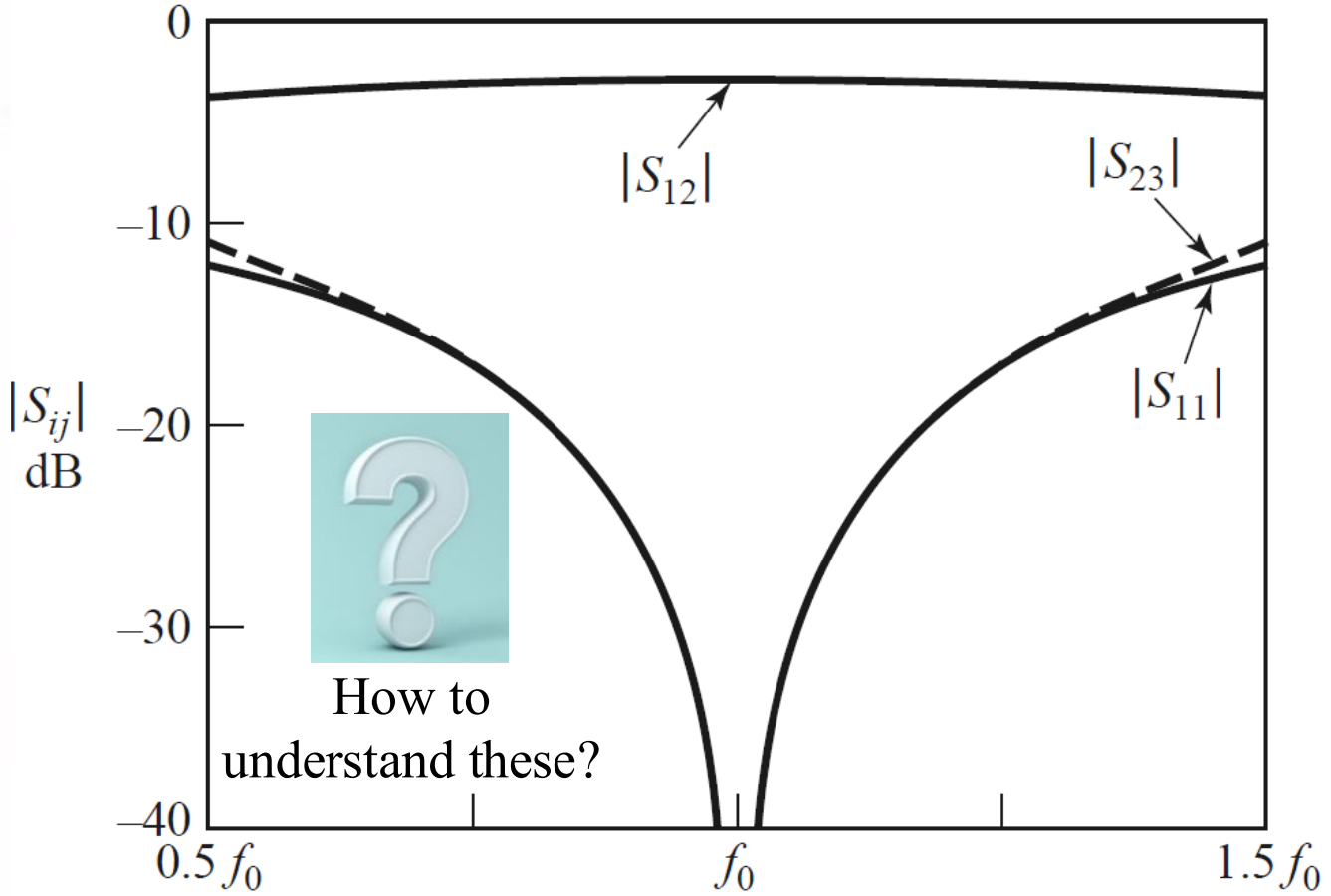
Power Divider

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Wilkinson divider: matched at all ports, reciprocal but lossy





equal-split Wilkinson power divider

Directional Coupler

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4 port network:

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix}$$

matched at all ports & reciprocal

$$\begin{bmatrix} 0 & S_{12} & S_{13} & S_{14} \\ S_{12} & 0 & S_{23} & S_{24} \\ S_{13} & S_{23} & 0 & S_{34} \\ S_{14} & S_{24} & S_{34} & 0 \end{bmatrix}$$

lossless

one solution

$$|S_{12}| = |S_{34}|$$
$$|S_{13}| = |S_{24}|$$
$$S_{23} = S_{14} = 0$$

choose $S_{12} = S_{34} = \alpha$, $S_{13} = \beta e^{j\theta}$, and $S_{24} = \beta e^{j\phi}$, α and β are real, and θ and ϕ are phase constants.
$\alpha^2 + \beta^2 = 1$, $\theta + \phi = \pi \pm 2n\pi$

symmetric coupler:

$$\begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$$

$\theta = \phi = \pi/2$

or

antisymmetric coupler:

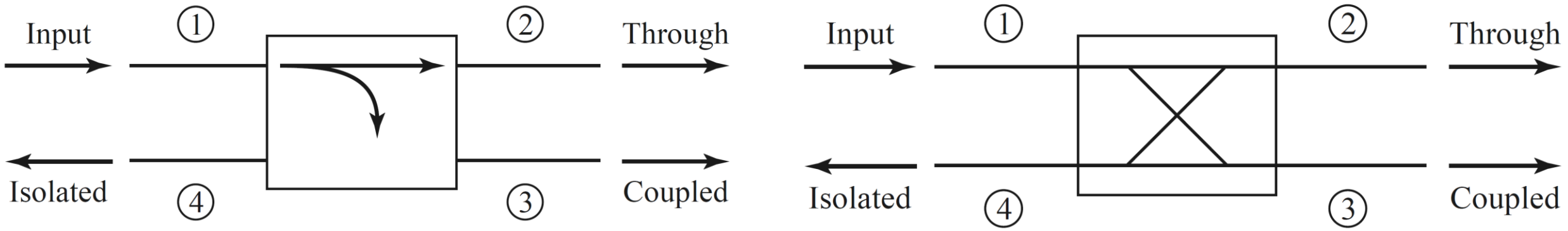
$$\begin{bmatrix} 0 & \alpha & \beta & 0 \\ \alpha & 0 & 0 & -\beta \\ \beta & 0 & 0 & \alpha \\ 0 & -\beta & \alpha & 0 \end{bmatrix}$$

$\theta = 0, \phi = \pi$

any reciprocal, lossless, matched four-port network is a **directional coupler**

Directional Coupler

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two commonly used symbols for directional couplers and power flow conventions

$$\text{Coupling} = C = 10\log\frac{P_1}{P_3} = -20\log\beta \text{ dB}$$

indicates the fraction of the input power that is coupled to the output port

$$\text{Directivity} = D = 10\log\frac{P_3}{P_4} = 20\log\frac{\beta}{|S_{14}|} \text{ dB}$$

a measure of the coupler's ability to isolate forward and backward waves (or the coupled and uncoupled ports)

$$\text{Insertion loss} = L = 10\log\frac{P_1}{P_2} = -20\log|S_{12}| \text{ dB}$$

accounts for the input power delivered to the through port, diminished by power delivered to the coupled and isolated ports

$$\text{Isolation} = I = 10\log\frac{P_1}{P_4} = -20\log|S_{14}| \text{ dB}$$

a measure of the power delivered to the uncoupled port

$$I = D + C \text{ dB}$$

Directional Coupler

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hybrid coupler (special case of directional coupler, coupling factor is 3 dB) S-parameters:

① $\alpha = \beta = 1/\sqrt{2}$, $\theta = \phi = \pi/2$



$$\frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 & j & 0 \\ 1 & 0 & 0 & j \\ j & 0 & 0 & 1 \\ 0 & j & 1 & 0 \end{bmatrix}$$

symmetric coupler:

quadrature hybrid 90° phase shift between ports 2 and 3, when fed at port 1

② $\alpha = \beta = 1/\sqrt{2}$, $\theta = 0, \phi = \pi$



$$\frac{1}{\sqrt{2}}\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

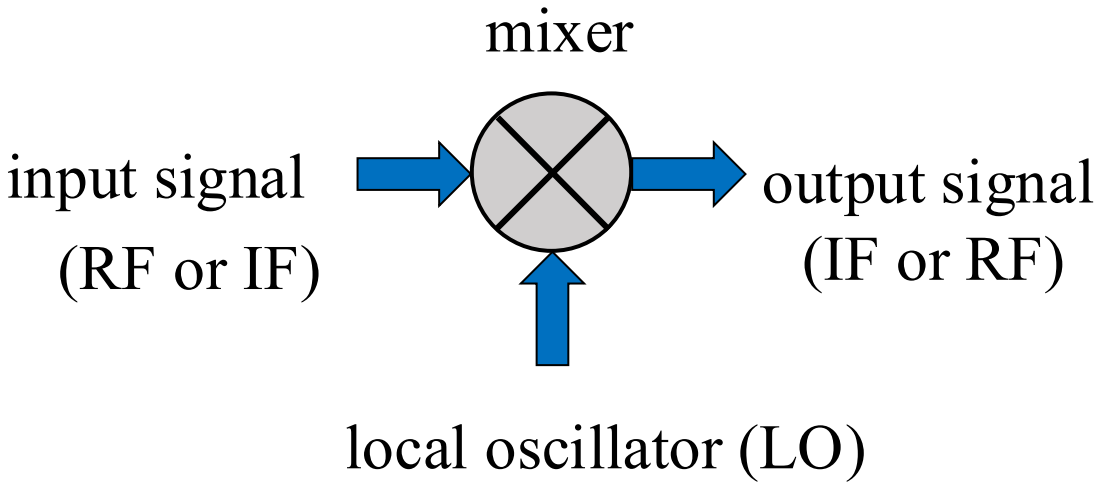
antisymmetric coupler:

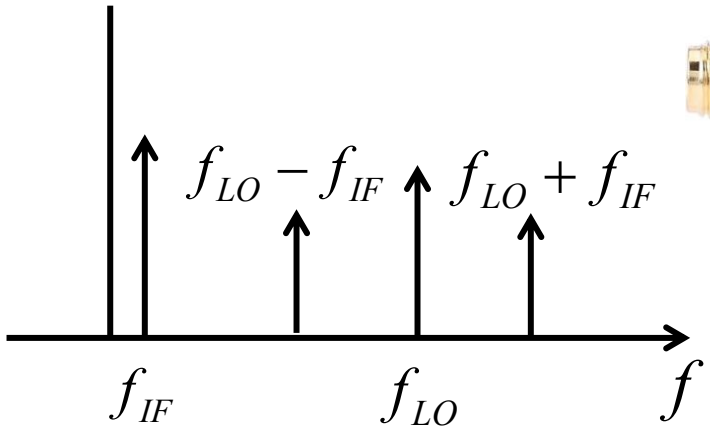
magic-T hybrid and rat-race hybrid 180° phase shift between ports 2 and 3, when fed at port 4

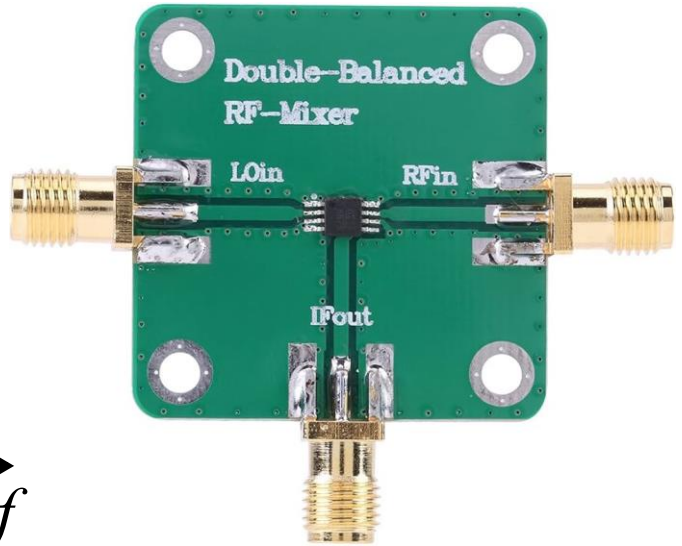
Mixer

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A mixer is a three-port device (two inputs and one output) that uses a nonlinear or time-varving element to achieve frequency conversion.

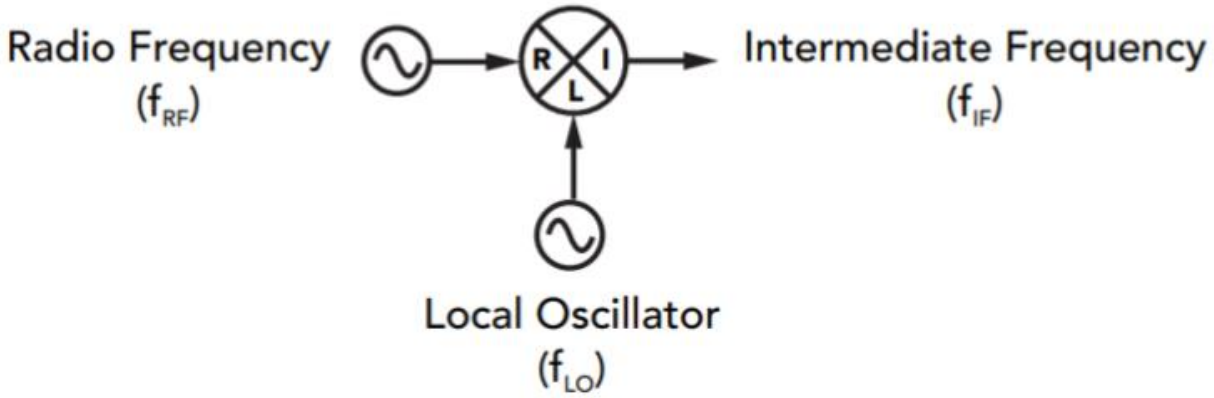






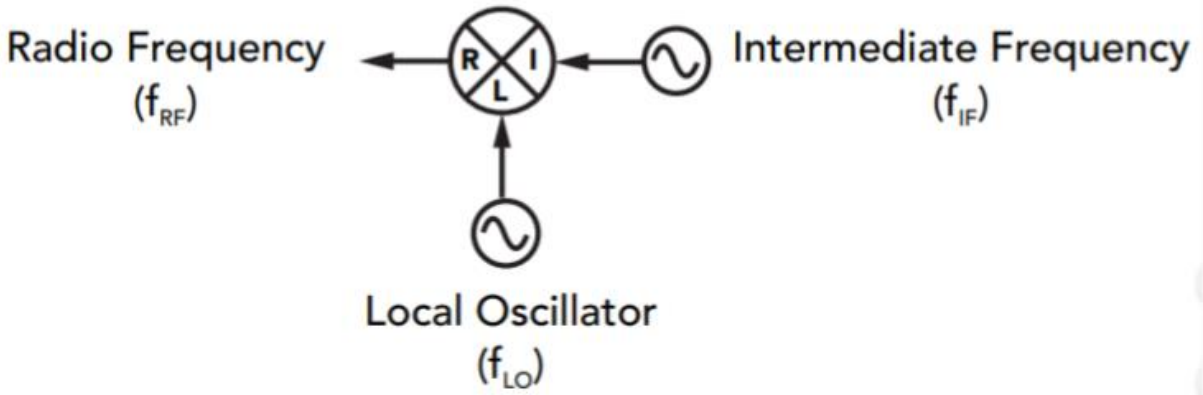
An ideal mixer produces an output consisting of the sum and difference frequencies of its two input signals. Operation of practical RF and microwave mixers is usually based on the nonlinearity provided by either a diode or a transistor.

Mixer



Using a down-conversion mixer in a receiver has several advantages:

- Different receivers can share the same IF channel for amplification and demodulation;
- Different receivers can use the same IF bandpass filter for channel selection;
- In higher frequency bands (millimeter or sub-millimeter bands), amplifier design is more difficult and expensive. Thus, converting the signal to a lower frequency allows you to apply manipulations on it more easily.

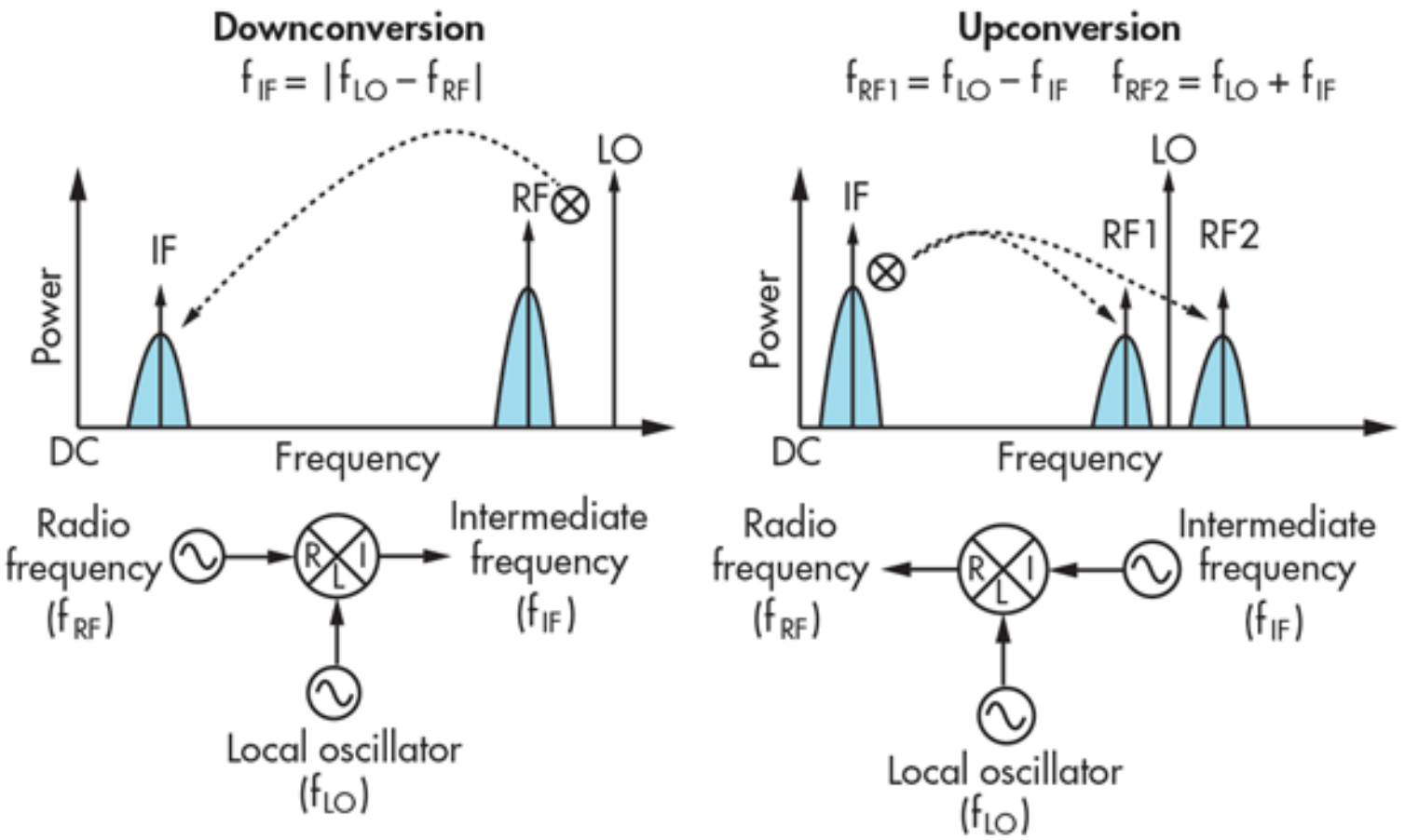


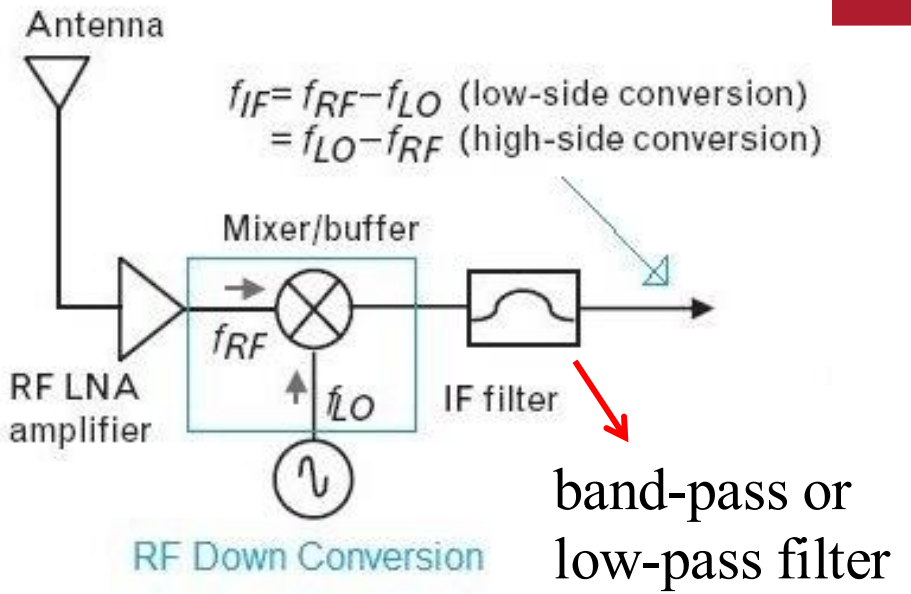
Using a up-conversion mixer in a receiver has several advantages:

- Transmitter and receiver can share the same local oscillator;
- It is easy to change the transmission channel by changing the frequency of the local oscillator;
- Using a subharmonic mixer, it is possible to transmit a high frequency with a low-frequency LO. (standard mixer: LO ± IF, sub-hamronic mixer: 2LO ± IF)

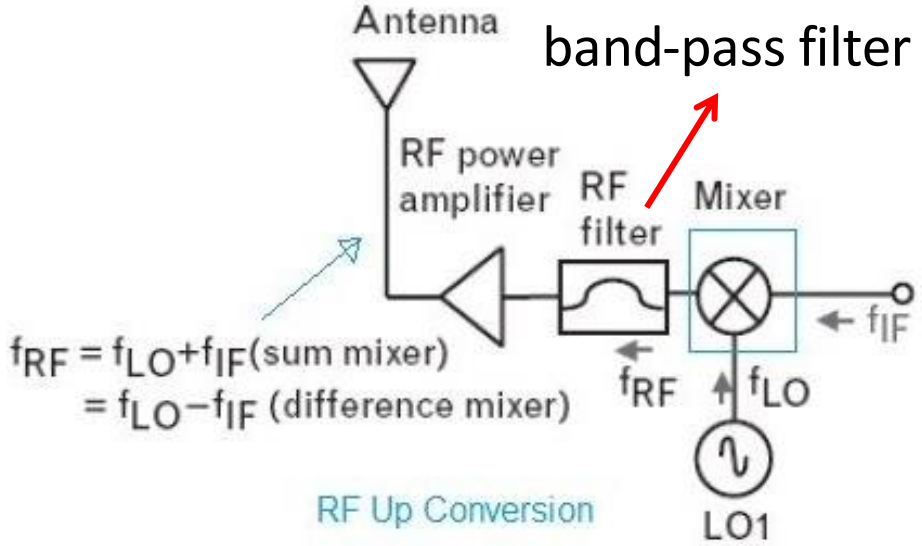
Mixer

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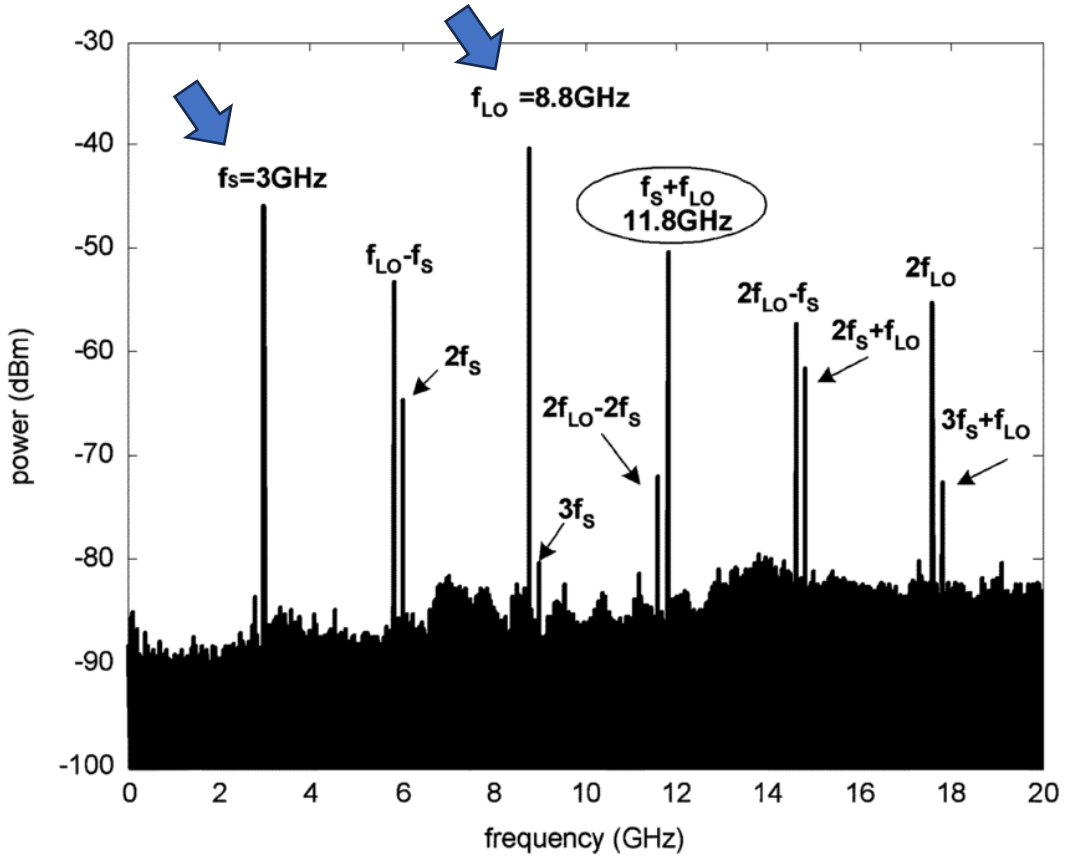
band-pass or low-pass filter



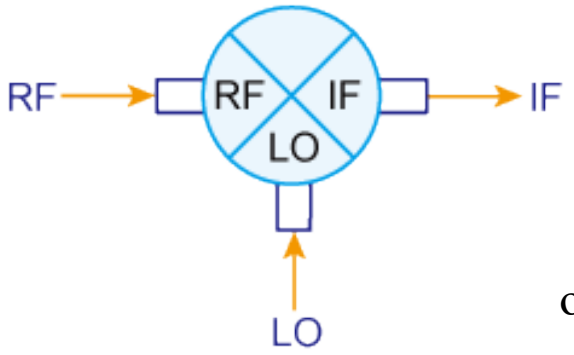
Mixer

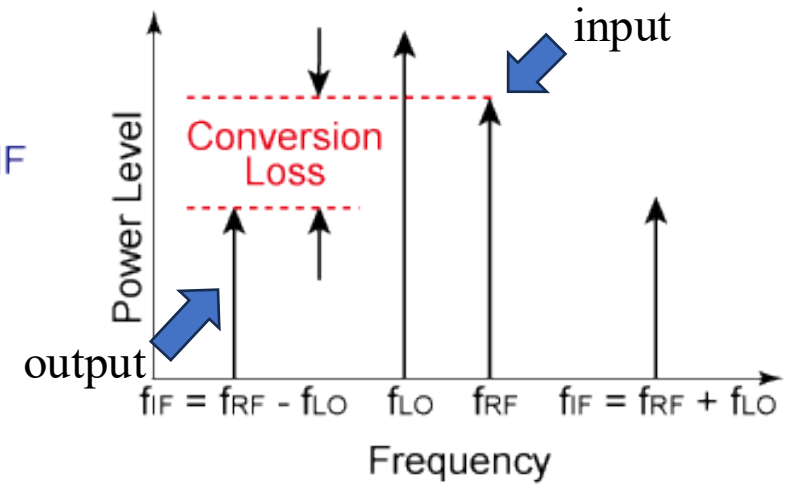
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Note that the above discussion only considers the sum and difference outputs as generated by multiplication of the input signals, whereas in a realistic mixer many more products will be generated due to the more complicated nonlinear behavior of the diode or transistor.



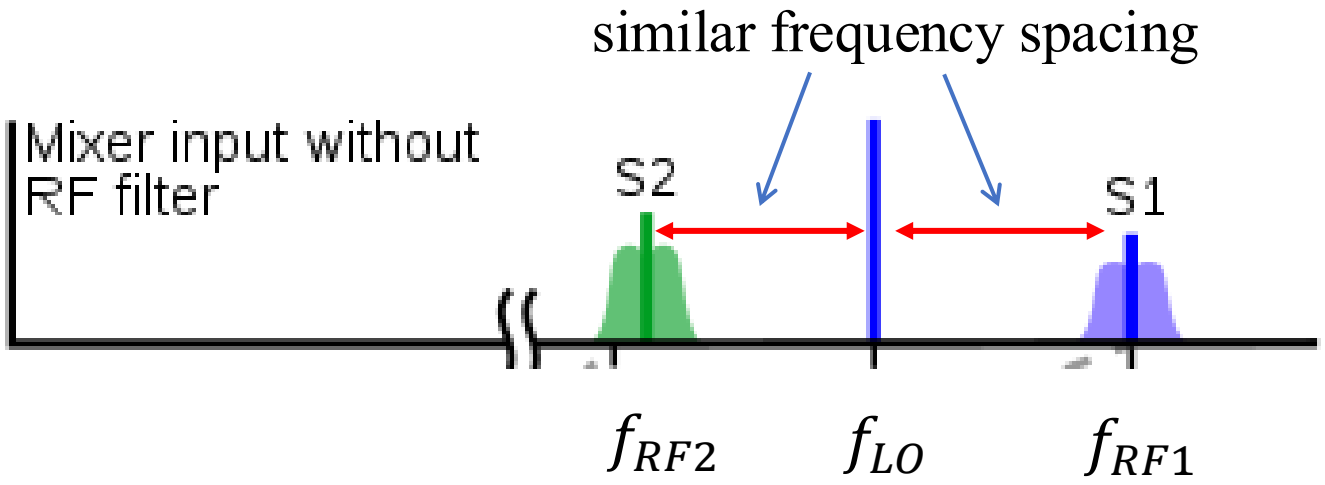
$$\text{Conversion loss [dB]} = 10 \log \frac{\text{Mag } (f_{IF})\ \text{mW}}{\text{Mag } (f_{RF})\ \text{mW}}$$





Mixer

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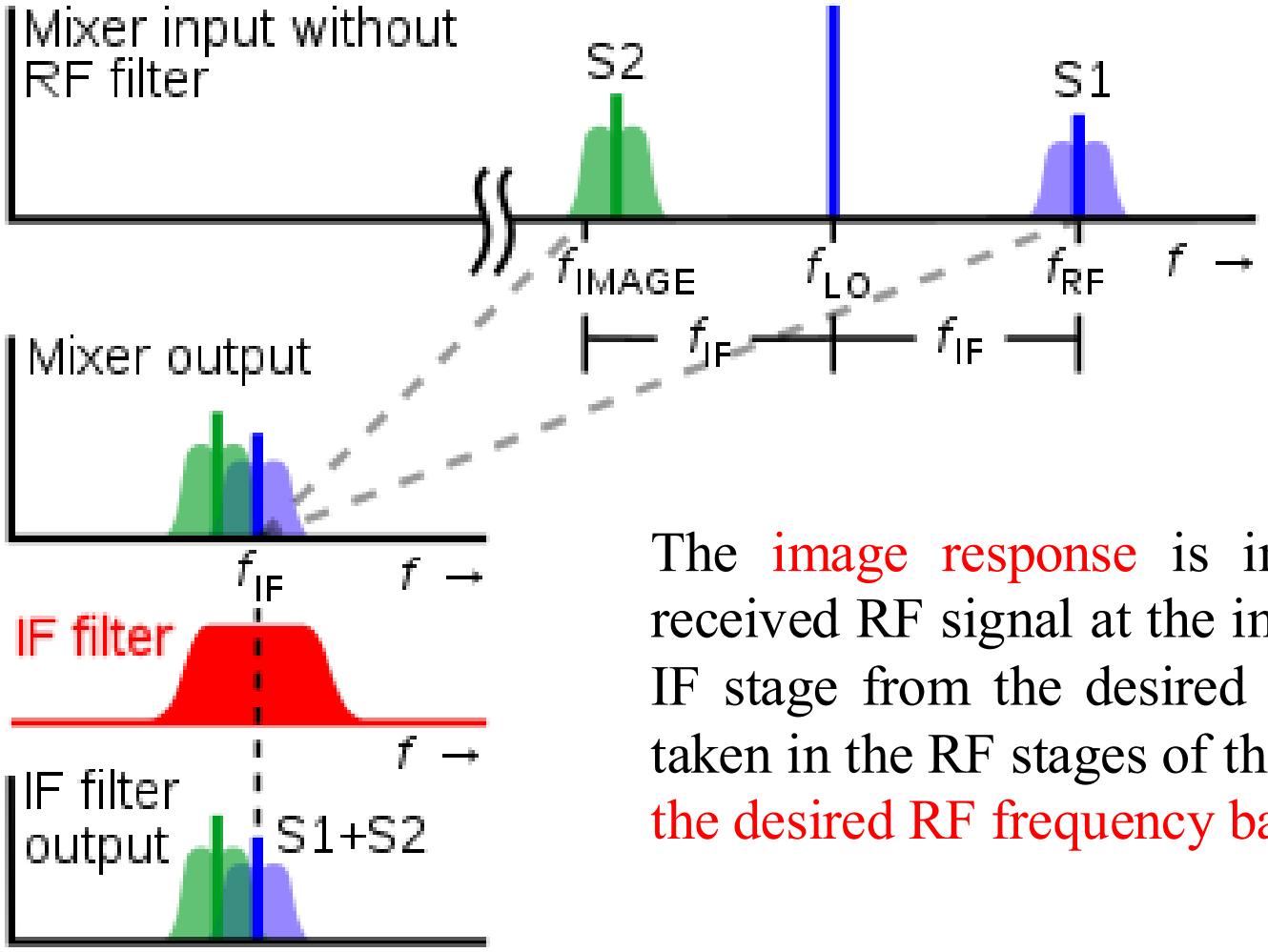




What happens after down-conversion in frequency domain?

Mixer

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What happens after down-conversion in frequency domain?

The image response is important in receiver design because a received RF signal at the image frequency is indistinguishable at the IF stage from the desired RF signal of frequency unless steps are taken in the RF stages of the receiver to preselect signals only within the desired RF frequency band.

Mixer

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In a receiver, the RF input signal at frequency f_{RF} is typically delivered from the antenna, which may receive RF signals over a relatively wide band of frequencies.

For a receiver, if RF input frequency is the sum of LO frequency f_{LO} and IF frequency f_{IF}, we have:

$$f_{RF} = f_{LO} + f_{IF}$$

After down-conversion (low-pass filtering), we would have f_{IF}. If the RF input frequency is:

$$f_{IM} = f_{LO} - f_{IF}$$ image response

We would have $-\boldsymbol{f_{IF}}$ after down-conversion (low-pass filtering). Mathematically, $\boldsymbol{f_{IF}}$ and $-\boldsymbol{f_{IF}}$ are identical as Fourier spectrum of any real signal is symmetric about zero frequency, including both negative and positive frequency.

Solutions: image-reject filter, image-reject mixer (specific design), and more

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Questions?

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